Building AI Applications with LLMs Using Python

From AI Fundamentals to Real-World AI Applications

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Build AI Applications the Right Way — Step by Step.

Introduction to Building AI Applications

- AI Stands for Artificial Intelligence
- AI is a broad field that includes many technologies and concepts.

- This diagram shows AI Ecosystem
- As you can see in this diagram, AI includes Machine Learning, Deep Learning, Large Language Models, RAG, AI Agents, Memory, Tools, Workflows, and many more.

- Don't worry... We're not going to learn everything at once. We'll learn one concept at a time.
- So, Where Should We Begin ?
    - we'll start with Large Language Models, because they are the foundation of most modern AI applications.

Introduction to Building AI Applications…

- **Software Application (OR) Traditional Software Application**
  - **Example: Simple Calculator Application**
    - **2 + 4 = 6**
    - **"What is the addition of 2 and 4?" or "What is the sum of two and four?" ==> No Answer**
    - **But as humans, we immediately understand the meaning.**
    - **Why ?**
      - **Because we have intelligence that helps us understand language and context.**

- **Now imagine if we could give that same ability to software.**
- **Of course, we cannot give it human intelligence.**
- **Instead, we provide Artificial Intelligence.**
- **Once an application has AI, it can understand natural language, interpret the user's request, and generate meaningful responses.**
- **Such applications are called AI applications.**
- **Where does this Artificial intelligence come from?**
- **Ans:** The answer is Large Language Models, or LLMs.
  - You can think of it as a machine's brain created artificially.
  - It understands natural language and generates intelligent responses.
  - Finally, LLM is an AI model/(brain) that provides intelligence to AI applications

- **Our Course is : Building AI Applications with LLMs**
- **What is an AI Application?**
  - **An AI application is a software application that uses Artificial Intelligence (AI) to perform tasks.**

Types of LLMs

- Many organizations have developed their own LLM's
- Some of the most popular LLMs include:
    - GPT (OpenAI)
    - Gemini (Google)
    - Claude (Anthropic)
    - DeepSeek (DeepSeek AI)

Key idea - Different companies have built different LLMs but they all have the same goal—to understand your request and generate a useful response.

- **Every LLM Has Two Important Limitations That Every AI Application Developer Should Understand Before Building AI Applications**
- **They are:**
    - **Knowledge Cutoff / Knowledge has a time limit**
    - **No Memory between conversations / Stateless conversions**

Your Brain

Stores knowledge from school, books, friends, experiences and life.

Machine Brain (LLM)

Trained on a vast amount of publicly available internet data.

- **Note: An LLM only knows information up to its training cutoff date and LLMs are stateless**

3 Key Ideas About LLMs To Build AI Applications

01

LLM = Machine Brain

Created artificially.
Learned from a vast amount of publicly available **internet data** — including websites, articles, books, forums, documents, and more.

DATA

02

Knowledge has a time limit

LLM only knows up to when it was built — just like your graduation year.

03

No memory between conversations

Every new conversation starts fresh.
We have to build memory ourselves.

Final Thought:

- Every powerful AI application starts with an LLM. Master the LLM, and the rest of the AI ecosystem becomes much easier to understand.